

GT Challenge Kit

WHAT'S IN THE KIT

- 4 Kanoodle games
- 4 3×3 Rubik's cubes
- 20 2×2 Rubik's cubes



QUICK-START GUIDE

1. Kanoodle: pick a challenge card, place the starting pieces, solve the rest.
2. Rubik's 2×2s are the friendly entry point — learn one solving method as a class.
3. Run as a puzzle station, early-finisher shelf, or full challenge day.
4. Track personal bests, not rankings.

FULL LESSONS IN THIS GUIDE

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PRO TIP: With 20 2×2 cubes, you can run a whole-class simultaneous solve — a great brain-break ritual.

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LESSON · GRADES K-2

Puzzle Persistence Center

LEARNING OUTCOMES

- Students will persist through challenges without immediate success
- Students will use trial-and-error strategically
- Students will reflect on how stuck feels and what to do about it

MATERIALS

- Kanoodle games (beginner challenge cards)
- Puzzle station setup with visual timer
- Feelings/strategy chart: 'When I'm stuck I can...'
- Progress sticker chart (attempts, not just solves)

INSTRUCTIONS

1. Introduce Kanoodle with the easiest card, modeling being stuck OUT LOUD: 'Hmm, that didn't fit. I'll turn it.'
2. Build the strategy chart together: turn the piece, start over, take a breath, look again.
3. Station time: students attempt cards at their level.
4. Sticker rule: attempts earn recognition, not just solutions — normalize the struggle.
5. Rotate weekly cards to keep the ladder climbing.
6. Circle share: 'Tell us about a time you were stuck and what you tried.'

ACTIVITY

Puzzle Persistence Center — a rotating station where 'stuck is where learning starts' is the posted motto. Stickers for attempts, not just solves, rewires what success means.

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LESSON · GRADES 3-5

Cube Club Method

LEARNING OUTCOMES

- Students will learn and execute a multi-step solving method
- Students will practice toward measurable personal improvement
- Students will coach a peer through a learned procedure

MATERIALS

- 2x2 Rubik's cubes (20 — enough for everyone)
- Beginner 2x2 method guide (printed steps)
- Personal best tracking sheet
- Timer per pair

INSTRUCTIONS

1. Whole-class simultaneous scramble — 20 cubes at once is an event.
2. Teach the beginner method one stage at a time: first face, then first layer, then top.
3. Practice pairs: one solves, one coaches from the printed steps.
4. Time trials begin only after first full solve — then it's personal bests, never rankings.
5. Coaching rotation: students who click early become station coaches.
6. Graph personal progress over the checkout window; celebrate biggest improvement, not fastest time.

ACTIVITY

Cube Club Method — the whole class learns a real solving algorithm, tracked by personal bests only. The coaching rotation turns early finishers into teachers instead of showoffs.

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LESSON · GRADES 6-8

Speedcube Statistics

LEARNING OUTCOMES

- Students will collect personal performance data across sessions
- Students will compute mean, median, and identify outliers
- Students will interpret an improvement curve

MATERIALS

- 2x2 and 3x3 cubes
- Solve-time log sheets (every solve, every session)
- Sheets/graph paper for analysis
- Method guides for reference

INSTRUCTIONS

1. Students learn or refresh the solving method (pair support as needed).
2. Data protocol: every timed solve gets logged — good, bad, and disastrous.
3. After 20+ solves: compute mean and median; spot the outliers.
4. The key question: which average best represents you, and why?
5. Plot solve times in sequence: the improvement curve appears.
6. Analysis write-up: describe your curve, explain the outliers, predict your next 10 solves.

ACTIVITY

Speedcube Statistics — students analyze their own improvement data, where the mean-vs-median debate actually matters (one disaster solve wrecks a mean). Statistics with personal stakes.

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LESSON · GRADES 9-12

Algorithm Analysis

LEARNING OUTCOMES

- Students will express a physical procedure in formal notation
- Students will analyze algorithm efficiency by move count
- Students will connect procedural notation to computational thinking

MATERIALS

- 3x3 cubes + 2x2 cubes
- Cube notation reference (R, U, F, and primes)
- Algorithm documentation template
- Move-count comparison sheet

INSTRUCTIONS

1. Teach standard cube notation: each face turn is a symbol; sequences are algorithms.
2. Students document one solving stage as pure notation — no prose allowed.
3. Trade documents: can a partner execute your algorithm exactly as written? Ambiguity fails.
4. Move-count analysis: compare two methods for the same stage — which is more efficient?
5. Discuss trade-offs: fewer moves vs. easier to memorize — a real CS tension.
6. Extension: research how speedcubers' algorithms are engineered and shared.

ACTIVITY

Algorithm Analysis — solving stages expressed in formal notation and analyzed for efficiency. The 'fewer moves vs. easier to learn' trade-off is the same one software engineers negotiate daily.